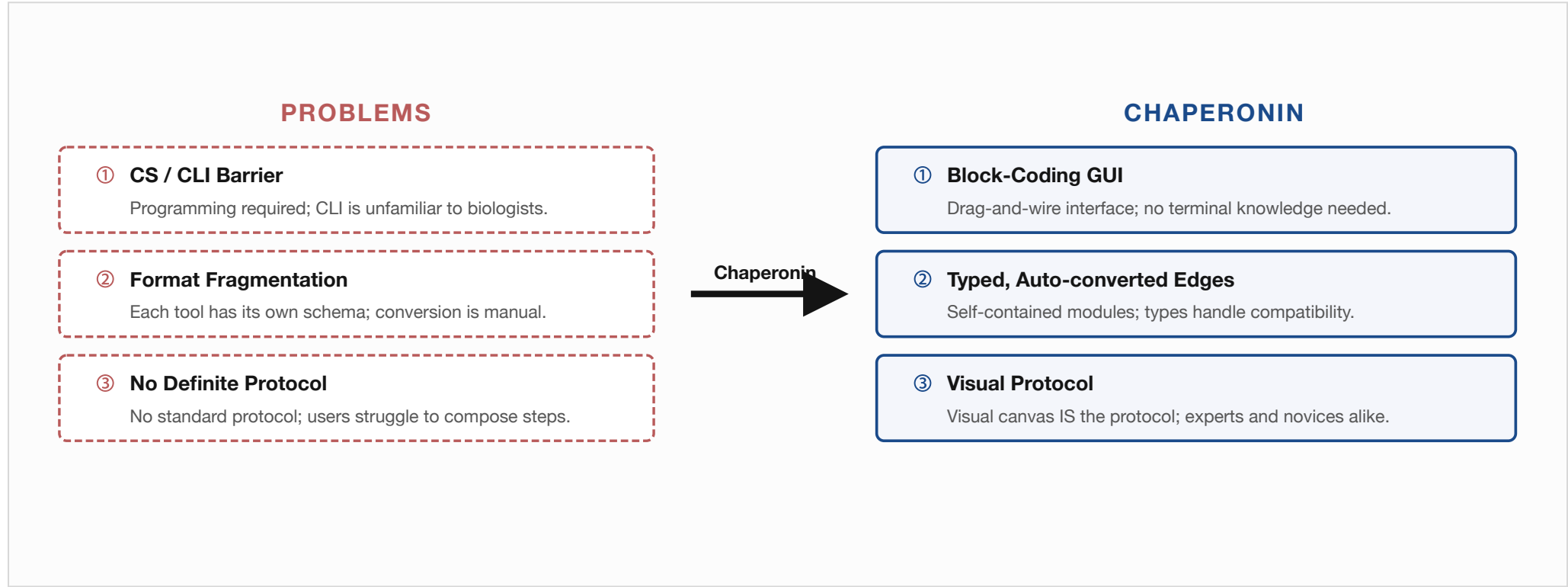


01 MOTIVATION

Biology first, no shell required.

Proteins are the key components in consistuting and sustaining an organism. Therefore, it is important to analyze and develop de novo (new) proteins to understand biological processes and develop new drugs. However, since the strcuture is highly complex, it is almost necessary to use various computational tools including AIs such as RFDiffusion, and AlphaFold. Unfortunately, those tools currently present some difficulties for biologists to utilize. First, they require computer science knowledge to operate, but compared to biological knowledge that biologists possess, they are lacking in computer competency, and thus often struggle with learning and using the tools. Also, most of them have CLI environments, which is counterintuitive for researchers to employ. Secondly, each tool is highly compartmentalized, requiring distinct file formats and data schemas, which eventually demands tedious manual data conversion processes. Lastly, researchers face significant challenges due to the lack of definite protein design protocol.

For these reasons, we developed a program that integrates a range of biological tools with a straightforward GUI. We were inspiired by block-coding, and designed the UI. Hence, users can effortlessly follow the protocol as required using it. Futhermore, each tool is developed and managed as a separate module, thus if any modification is needed, it can be done by editing each module rather than the whole program. This architecture also allows easy expansion to support other tools by adding new modules. Additionally, since the platform utilize actual tools from protein research, it maintains professionalism. Thus, researcher can combine their biological expertise with this program to conduct research in depth, while students or beginners can engage in biological reseaches without difficulties.



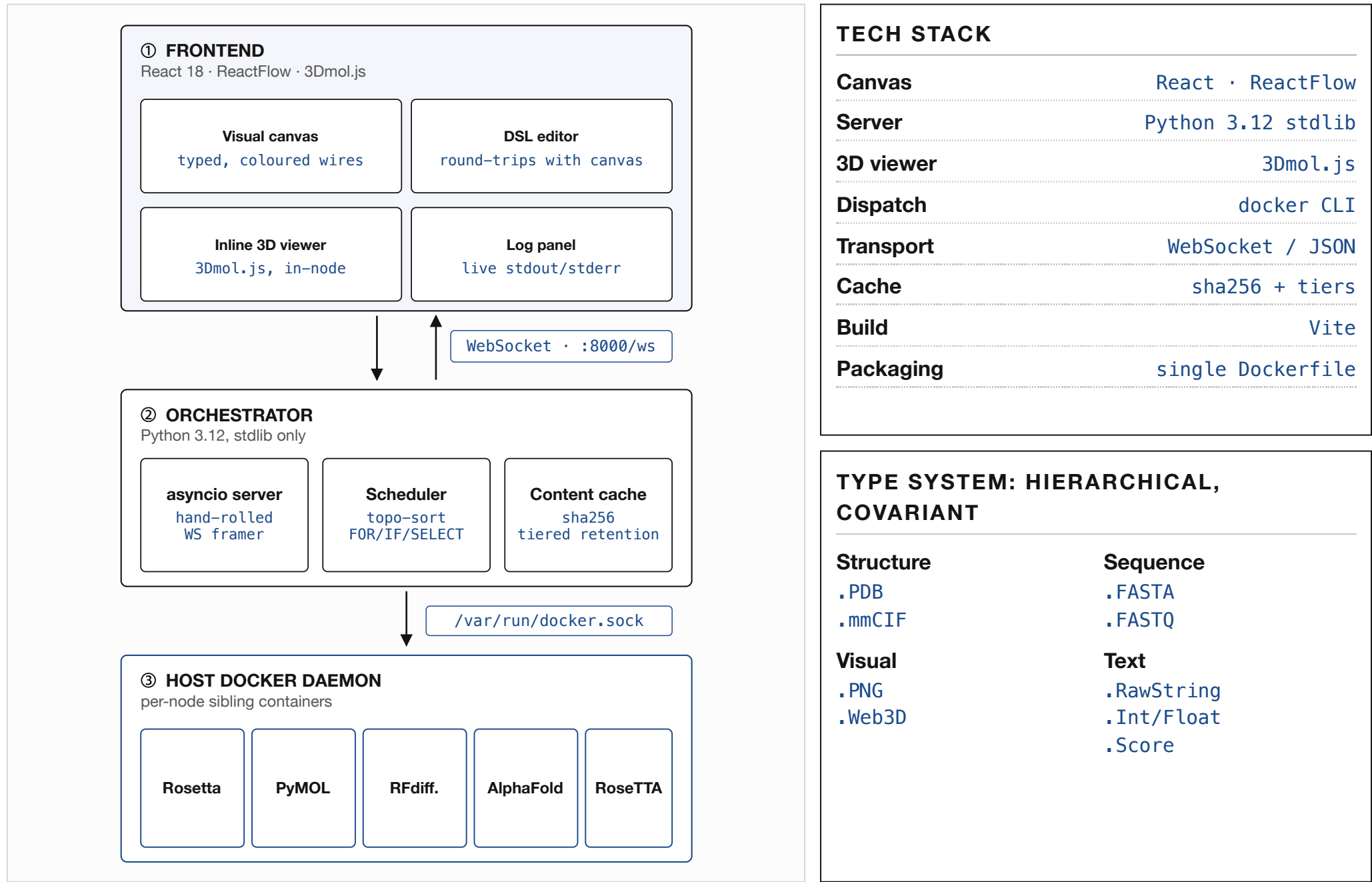
02 WORKFLOW

Five steps, no shell.

01 DRAG	Pick a module from the categorised palette (design, refinement, prediction, visualisation, converters).
02 WIRE	Connect typed handles. Edges are coloured by data type; incompatible wires are refused.
03 VALIDATE	Type-checking and dangling-input detection run at edit time, before any GPU-hour is spent.
04 RUN	Scheduler topo-sorts, dispatches each node as a sibling Docker container; cache short-circuits hits.
05 STREAM	Logs, progress bars and 3D structures stream live into the canvas. Stop kills containers instantly.

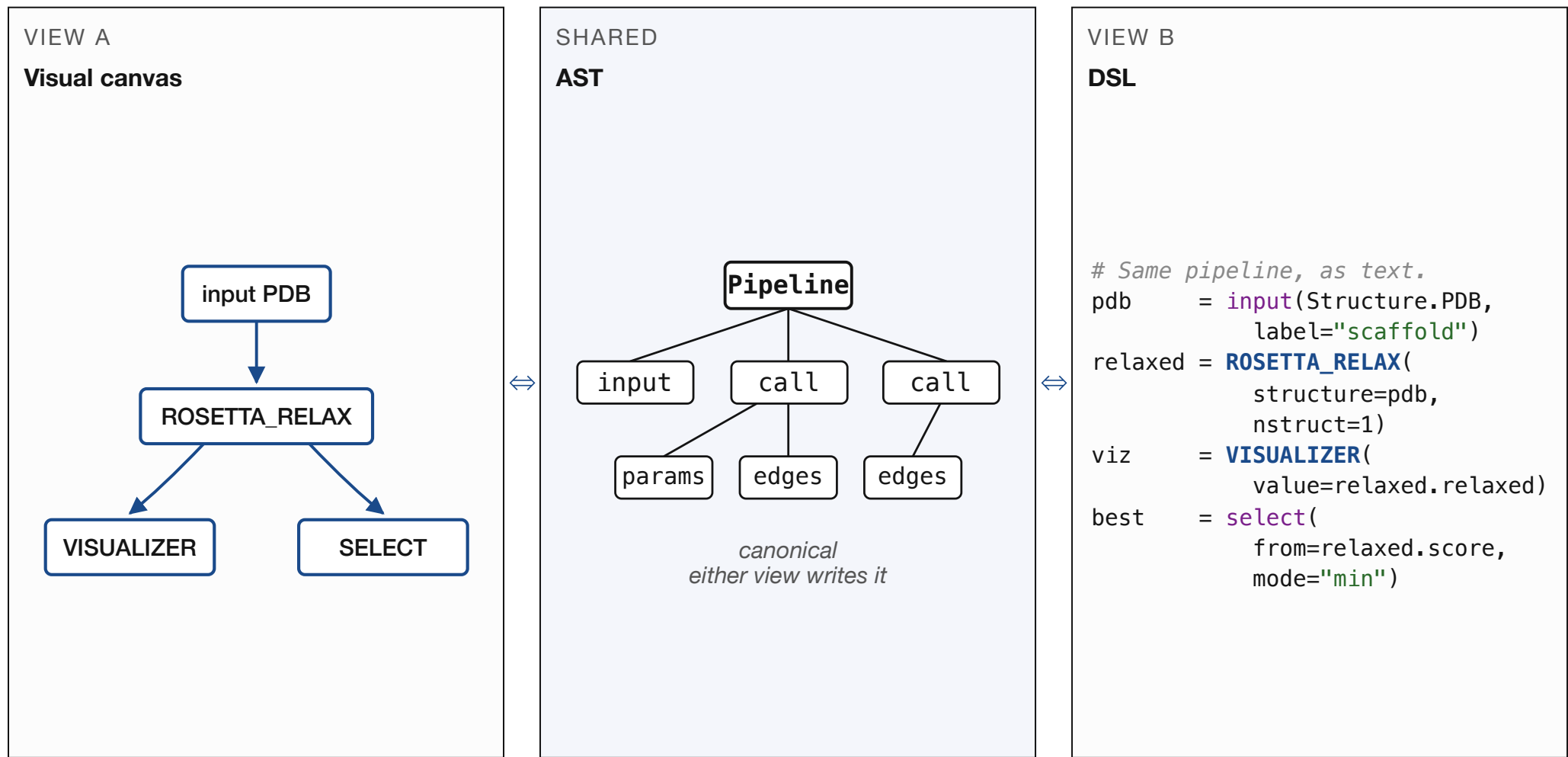
03 ARCHITECTURE

Four layers, one socket.



04 TRANSLATION

Two views, one AST.

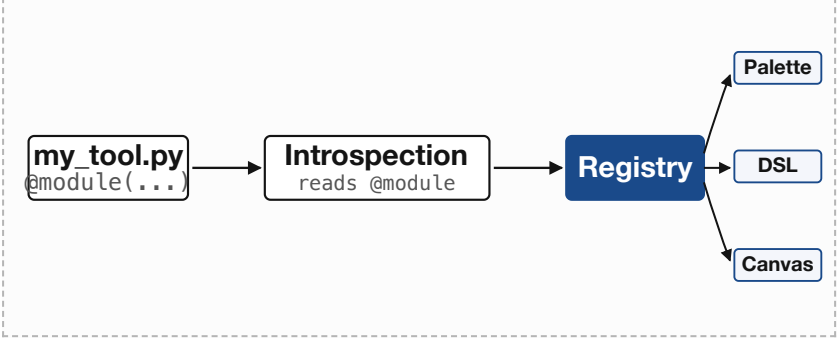


05 EXTENDABILITY

One Python file, one module.

```
# backend/modules/rfdiffusion.py
from chaperonin import module, Input, Param, Output
from chaperonin.types import Structure, Text
@module(name="RFDIFFUSION", category="design",
  container="ghcr.io/rosettacommons/rfdiffusion:1.5.0",
  resources={"gpu": 1, "memory_gb": 24})
class RFDiffusion:
    pdb_file: Input[Structure.PDB]
    hotspot: Param[Text.RawString]
    designed_pdb: Output[Structure.PDB]
    def execute(self, ctx):
        ctx.run(["rfdiffusion", ...])
        ctx.publish("designed_pdb", ctx.workdir / "designed.pdb")
```

Drop a single .py file under backend/modules/. The @module decorator self-describes inputs, params, outputs, resources and container image. No registry edits, no scheduler patches.



06 RESULTS

End-to-end on a laptop.

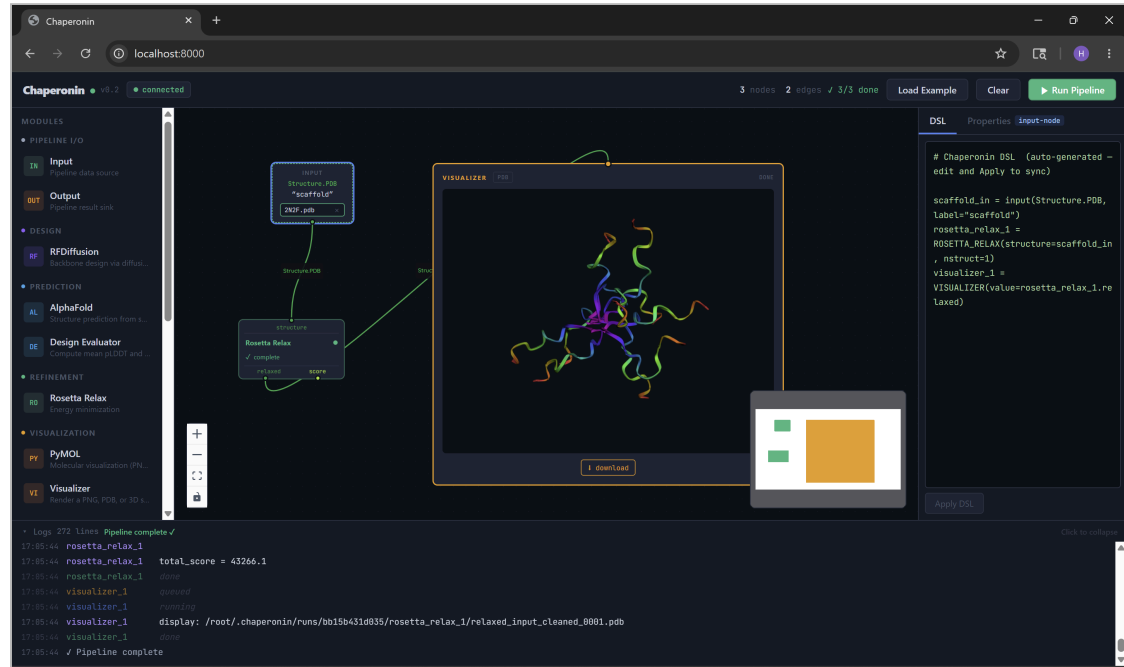


Fig. 1 Rosetta Relax Pipeline.

The relaxations step in crucial preprocessing procedure required to run protein design tools such as RoseetaFold. As illustrated, the pipeline receives a portein structure file (in a format of .pdb) as input, processes it through the Rosetta Relax module, and renders the relaxed struture via the Visualizer module. From the user's perspective, the entire process is reduced to connecting three nodes on a visual canvas without any command line (CLI) interaction. The relaxed structure is rendered directly within the program, eliminating the requirement for external visualization tools.

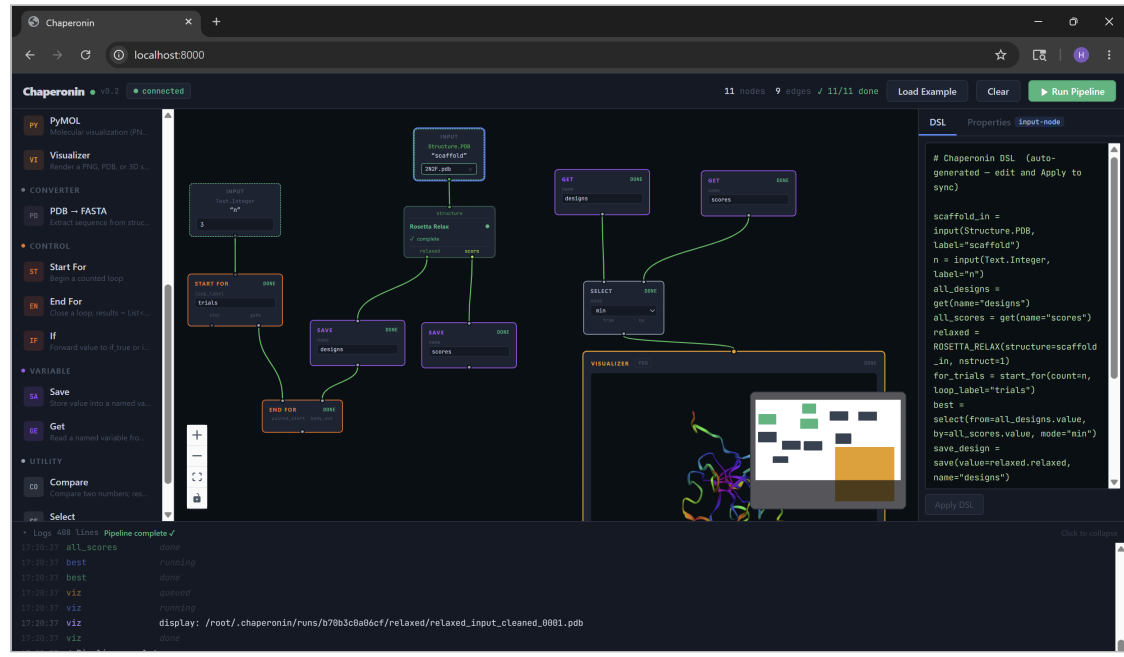


Fig. 2 Iterative Execution & Conditional Logic.

The second result demonstrates the function for iterative execution and conditional logic within a pipeline. As shown, the program runs the Rosetta Relax module three times and selects the output with the lowest energy score. Thus we can automatically choose the most stable structure with the Select node. Not only Rosetta Relax, but all other tools can equally be incorporated into loop and conditional structures through the Start For, End For and Select nodes. By allowing multiple trials and conditional selection within a pipeline, researchers can systematically search a range of outputs and identify the optimal result.

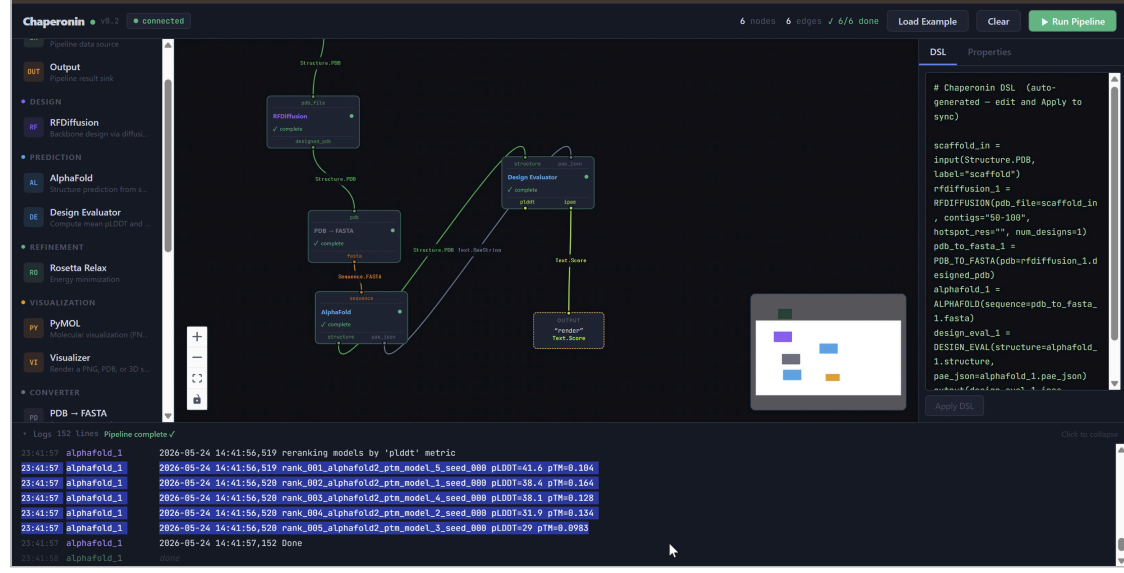


Fig. 3 De Novo Protein Design Pipeline.

The third result shows a complete de novo (new) protein design pipeline, analogous to a real-world drug development workflow. As shown in the figure, a target protein structure is first passes via RFDiffusion module and generate a binding protein. This de novo protein next moves to the AlphaFold module and evaluated by the iPAE scores, which represents how accurately the position is predicted. This workflow is directly mirrors the process of developing a binding protein in drug discovery.

07 TESTIMONIALS

From the field.

Being able to connect different computational programs and run them on a single platform would be enormously convenient. If Chaperonin really lets a researcher wire the tools together with a mouse and execute them, it will be an exceptionally practical bioinformatics tool.

PROF. AN JEONG-HUN · Dept. of Chemistry & Biology

08 CONCLUSION

Typed pipelines, fewer glue scripts.

In conclusion, Chaperonin addresses the key difficulties that many biologist face in protein design. By integrating established tools such as RFDiffusion, Alphafold, Evaluator, Visualizer and various other tools, within an intuitive node-based GUI, this program offers highly effective environment for both researchers and beginners who waish to design their own de novo proteins. Futhermore, by incorporating loop and conditonal structures into the pipeline, the entire processes can be conducted systemmatically. With this platform, users can easily follow the workflows directly analogous to the real-world research, including drug development. Futher work will focus on adding new modules such as RosettaFold to improve its functions and versatility.